

Response to Reviews of “Detecting sparse biological invasions through multi-scale remote sensing: a drone–satellite framework for mapping Neltuma in dryland ecosystems”

Editor

Dear Editor,

We thank you for considering our manuscript and are grateful to both reviewers for their positive assessments and thoughtful, constructive feedback. Their comments have helped us to substantially improve the clarity, rigour and presentation of the manuscript.

We have carefully considered every comment and have revised the manuscript extensively in response. Where appropriate, we have incorporated the reviewers' suggestions directly into the manuscript through additional methodological clarification, strengthened interpretation, integration of further relevant literature, and refinement of the discussion and conclusions. Our objective has been to present a robust evidence base that informs future methodological development and supports researchers and practitioners in making context-specific decisions across different applications, geographic settings and Earth observation technologies, rather than prescribing a single universally applicable workflow. In a small number of cases, we have respectfully retained our original approach where the suggestions would have required a fundamental redesign of the study or addressed research objectives beyond the scope of the work presented. In these instances, we have sought to clarify our rationale while ensuring that the manuscript more clearly communicates the strengths, limitations, and broader applicability of the framework.

We believe the revised manuscript is substantially stronger as a result of the review process. Our point-by-point responses are provided below. Reviewers' comments are shown in black, our responses in blue, and quoted additions to the manuscript are shown in italics. All line numbers refer to the revised manuscript without tracked changes.

Thank you again for your time and consideration. We look forward to hearing from you.

Yours sincerely,

Glenn Slade and Andy Cunliffe

To pick up with Hugh

Following an review of the codebase, we have lower confidence than I would like that the implemented workflow fully matches the manuscript narrative. There do not appear to be obvious coding errors, but there are several areas where the analytical implementation and manuscript descriptions diverge. In particular, we should verify exactly how the spatial blocking operates across the 200 iterations, what uncertainty is actually quantified, and whether our claims around "robust uncertainty estimation" are fully supported by the implemented analyses. More broadly, the analysis would have been considerably strengthened by probabilistic prediction outputs rather than hard classifications, with explicit uncertainty mapping. Looking ahead, conformal prediction would be an excellent addition, providing calibrated predictive uncertainty with relatively little change to the overall modelling framework. **It might be valuable for us to work through the pipeline to ensure the manuscript accurately reflects the code and to identify any opportunities for strengthening the final analyses in light of the (valid) reviewer feedback.** This will help elevate this from "just another great paper" towards "landmark of how this kind of mapping should be undertaken"

Next steps for codebase cleanup

Order Person Action

- | Order | Person | Action |
|-------|--------|--|
| 1 | Hugh | Implement targets across the analytical workflow, including explicit dependencies between data preparation (pulling most from Zenodo), modelling, validation and outputs. Reproducible, auditable workflow and clear provenance of manuscript outputs. |
| 2 | Hugh | Standardise class names.
Convert Neltuma_Mlr3_Pipeline/data_in/Veg_type_lookup_list.xlsx to json |
| 3 | Hugh | Change to analysis centred class probabilities. restructuring the prediction stage to retain class probabilities (predict_type = "prob" where supported) as a foundation for future conformal prediction. |
| 4 | Hugh | Resolve the SPCV iteration discrepancy. Establish exactly how many spatial resampling iterations were used for the reported results, including how the seven spatial blocks relate to the repeated resampling. Remove redundant/conflicting SPCV objects. Audit the spatial blocking implementation. Document which observations/AOIs can occur in each training/test partition and confirm that spatial separation is implemented as described in Sections 2.5–2.6 or update description. reconcile the apparent 100 vs 200 iteration discrepancy. |
| 5 | Hugh | Render outputs (Manuscript and Supplementary Information) as Quarto in Word |
| 6 | Hugh | Audit prediction outputs and uncertainty. Establish exactly what is currently generated: hard classifications, class probabilities, validation distributions, etc. Confirm precisely what can legitimately be described as "uncertainty assessment". Other things to consider: Hyperparameter tuning is relatively lightweight (term_evals = 5). The manuscript does not describe this in detail. Not necessarily a problem, but worth explicitly documenting to improve reproducibility. |

Order Person Action

- 7 Hugh **Check all key reported statistics against the code/output.** In particular, verify overall/class accuracies, 24.8% satellite overprediction, 34.6% potential misclassification, sample sizes and resampling counts.
- 8 Hugh **Add renv and capture the computational environment.** Initialise the project, install/use the required package versions and generate renv.lock.
- 9 Hugh **Improve repository documentation.** Add a concise workflow description/README showing execution order, key inputs, generated outputs and dependencies.
- 10 Hugh **Optional? Add basic reproducibility checks.** Where practical, create a small validation step that checks key outputs against expected manuscript statistics, such as sample size, number of resamples and headline accuracy metrics.
- 11 Hugh + Andy **Consider whether a probabilistic/conformal analysis is warranted.** Treat this as an analytical extension, not part of basic code cleanup.
- 12 Andy **Reconcile the revised manuscript with the audited codebase.** Review Methods, Results, Discussion, Abstract and Conclusions against the confirmed implementation, particularly SPCV, uncertainty, class structure and sample size.
- 13 Andy **Reconcile the Supplementary Information.** Check Tables S3/S4 and all methodological/validation descriptions against the final code and outputs.
- 14 Andy **Reconcile the responses to reviewers.** Once the code and manuscript are settled, update all responses so that claims about the analysis, validation and uncertainty are fully supported by the final implementation.

Reviewer 1

The manuscript is generally well written, with strong visual presentation and a study that addresses an important challenge in the remote sensing of sparse invasive species. The inclusion of multiple imagery sources, varying spatial resolutions, and both internal and external validation approaches is a particular strength of the work. The manuscript would, however, benefit from major revisions including greater methodological transparency regarding the study design, **uncertainty assessment**, validation framework, and the links between the methods and reported results. Several analytical decisions would benefit from additional justification, and some results require clearer interpretation and contextualisation. The broader implications of the approach could also be strengthened through a more explicit discussion of transferability, practical applicability, and the trade-offs associated with incorporating drone-derived data into operational mapping workflows. In addition, there are a number of technical corrections that should be addressed.

Thank you for this constructive and encouraging assessment. We are grateful for the reviewer's detailed suggestions, which have helped us substantially strengthen the manuscript. In particular, we have improved the transparency of the study design and methodological framework, clarified our treatment of uncertainty and validation, strengthened the justification for key analytical decisions, and improved the interpretation and contextualisation of the results. We have also substantially revised the Discussion to better articulate the broader relevance and transferability of the approach, practical considerations around integrating fine-resolution UAV observations into operational mapping workflows, and the associated trade-offs. We believe these revisions have substantially improved the manuscript and made the study clearer and more useful for readers seeking to apply or extend multi-scale Earth observation approaches to sparse ecological targets.

Abstract

This sentence in the abstract seems to be AI-generated: *"Spatially explicit validation yielded lower but more realistic predictive performance than conventional approaches, underscoring the importance of robust uncertainty assessment."* Is very dense and yet vague. Did authors explicitly test for spatial autocorrelation? What do you mean by "uncertainty assessment"? Please write this in simple, clear English.

Thank you for highlighting this unclear sentence, we agree that the original wording was insufficiently specific. We have replaced it with: *"Models evaluated using spatial block cross-validation produced lower, but likely more realistic, accuracy estimates than random cross-validation, highlighting the importance of accounting for spatial dependence when assessing model performance."* This revised wording more clearly describes the validation approach we used and explicitly identifies the comparison with random cross-validation. We did not conduct a formal test of spatial autocorrelation; rather, spatial block cross-validation was used to account for spatial dependence among samples during model training and evaluation.

Why does the abstract not make recommendations as to satellite imagery selection? I feel that this is missing. Also the result on the best classifier is missing. Perhaps this

section: "Beyond species mapping, our results show how integrating fine-resolution ecological observations with satellite Earth observation can help overcome scale mismatch, improving early detection and management of sparse biological invasions. This multi-scale framework has wider relevance for biodiversity surveillance, woody encroachment, degradation monitoring and dryland early-warning systems." Can be replaced by more specific recommendations.

Thank you for this suggestion. We agree that practical recommendations are important and have revised the final part of the Abstract to better articulate the broader implications of our findings. We now conclude with: *"Beyond species mapping, our results show how integrating fine-resolution ecological observations with satellite Earth observation **via a hierarchical drone–satellite remote sensing framework** can help overcome scale mismatch, improving early detection and management of sparse biological invasions. This multi-scale framework has wider relevance for biodiversity surveillance, woody encroachment, degradation monitoring and dryland early-warning systems."* We believe this better reflects the principal contribution of the study: moving beyond over a decade of drone being proposed as a potential solution to demonstrating how they can be integrated within a hierarchical Earth observation framework to bridge scale gaps and strengthen landscape-scale ecological monitoring.

Regarding the suggestion to make prescriptive recommendations on satellite imagery selection, we respectfully disagree. Our study was designed to demonstrate how sensor grain influences the detectability of sparse *Neltuma*, rather than to identify a universal threshold at which a particular sensor becomes suitable or unsuitable for operational mapping. The most appropriate imagery will depend on the spatial characteristics of the target species (e.g. canopy size, density and spatial configuration), the intended management application, and practical considerations such as cost and data availability. We therefore consider that making a general recommendation in favour of a particular sensor would overstate the scope of our findings. Instead, the Abstract already communicates the principal practical result by explicitly quantifying the relationship between sensor grain and detection performance: *"Detection accuracy increased strongly with finer sensor grain, from ~50% using Sentinel-2 (10 m), to ~62% using PlanetScope (3 m) and ~77% using WorldView-2 (1.6 m), identifying scale mismatch as a major constraint on observing sparse ecological targets."* This provides readers with the evidence needed to judge whether a given level of performance is appropriate for their own ecological targets and management objectives, without making recommendations that extend beyond the available evidence.

Regarding the suggestion to emphasise the highest-performing classifier in the Abstract, we have chosen not to do so because identifying an optimal machine-learning algorithm was not a primary objective of this study. Rather, comparing multiple algorithms formed part of our methodological framework to ensure that the reported findings were robust to classifier choice. Although SVM and the ensemble meta learner generally performed best, the differences among algorithms were relatively modest compared with the much stronger effects of sensor grain and cross-scale calibration, which represent the principal scientific contributions of the study. We therefore consider it more appropriate for the Abstract to focus on these broader ecological and remote sensing insights than on the performance of a specific classifier.

Introduction

L62: "For example, the BORAVAST Community Trust Charcoal Project in our study area proved unsustainable in part because management resources were insufficient to address the scale of invasion." -> doesn't connect to the prior sentence. The prior sentence is about the challenge to map density and occurrence of these invasive species. But "insufficient to address scale" makes it seem that there was too much of the wood for the project to cope, when I presume the actual issue is that there are large areas with low densities, and small areas with high densities, meaning that implementation of charcoal projects (e.g. transporting kilns etc) rapidly become challenging. Language needs to be clarified.

Thank you for highlighting this disconnect in our narrative. We agree that the original mentions of the BORAVAST project in the introduction did not clearly explain the underlying management challenge and could be misinterpreted as simply reflecting insufficient resources to address the overall extent of the invasion. We have therefore removed this example from the Introduction and revised Section 4.4 to more clearly explain how the widespread distribution of low-density infestations, interspersed with smaller areas of high-density invasion, creates substantial logistical and economic challenges for utilisation-based management, particularly through increased harvesting, transport and processing costs. We have also strengthened the discussion of the implications of this spatial configuration for the feasibility of landscape-scale management and the need for spatially differentiated intervention strategies.

L78: also see Holden et al. 2020 -Mapping invasive alien trees in water towers -A combined approach; this study also incorporated drones, though not in the same way.

Thank you for highlighting Holden et al. (2021). We were already aware of this valuable study and agree that it provides a relevant example of integrating drone and satellite observations for invasive tree mapping, albeit using a different approach to that adopted here. We did not feel that a fifth citation was necessary to support the general statement in the Introduction that drone-based remote sensing can capture ecological structure at resolutions inaccessible to satellites and provide an intermediary between field observations and landscape-scale Earth observation, given the multiple citations already supporting this point. However, we have now incorporated Holden et al. (2020), alongside several additional relevant studies, into the revised Discussion, particularly Section 4.2, to better situate our findings within the broader multi-scale Earth observation literature and to emphasise how UAV surveys can be strategically targeted to locations where additional fine-resolution observations are likely to provide greatest value for calibration and uncertainty reduction, rather than requiring systematic landscape-scale coverage.

This paper may also be of interest: Skosana et al. 2025 -Exploring the tradeoffs between spatial and spectral resolution in mapping.

Thank you for signposting us to Skosana et al. (2025). We have read the paper and, following the references therein, have also reviewed several of the underlying sources, particularly the Foxcroft et al. material. These sources helped inform our revisions to Section 4.4, particularly by strengthening our discussion of the practical and collaborative dimensions of invasive species management, although we do not cite Skosana et al. (2025) directly in the revised manuscript.

Methods

L134: what is image A, give credits. Is it a true colour S2 scene?

Thank you for this comment. Figure 1D (not Figure 1A) is a true-colour Sentinel-2 scene. The image type and source are provided in the original figure legend.

L146: we need a map of these ground observations and more details. What was the sample size, what was the spread? If you only collected training data within drone footprints, then you would need to be especially careful of spatial autocorrelation. Was uncertainty assessment done? Did you produce an uncertainty layer? If not, what is meant in the abstract?

Thank you for this important point. We clarify that all training observations were collected within the seven drone survey areas, as described in Section 2.2. The field sampling design comprised 1,024 training observations distributed across the seven drone survey areas, with a minimum target of 20 observations per land-cover class within each area. To improve transparency, we have added Table S4 ("Number of training observations used for machine learning classification by land-cover class and drone survey area"), which summarises the sample sizes by class and survey area. Section 2.2 also describes the independent plant-scale validation using 184 Neltuma individuals across two 100 × 200 m plots, while Section 2.3 explains how the seven drone survey areas were distributed across gradients in Neltuma density, geomorphology and vegetation structure. The spatial distribution of the drone survey areas is shown in Figure 1D.

We agree that collecting training observations within the drone survey areas necessitates careful consideration of spatial dependence. Rather than undertaking a formal test of spatial autocorrelation or producing a spatial uncertainty layer, we addressed this through the repeated block-based spatial cross-validation framework described in Sections 2.5 and 2.6. Models were repeatedly evaluated using spatially distinct training and test partitions, providing a more conservative assessment of predictive performance and reducing the risk of overly optimistic accuracy estimates arising from spatially structured observations. We have clarified this distinction in the revised Methods and Discussion and have also revised the Abstract to avoid implying that a spatially explicit uncertainty layer was produced.

L202-203: please create a visual to show what is meant.

Thank you for this suggestion. We agree that visualizing differences among sensor spatial grains can be helpful for readers. However, we have elected not to add an additional figure because this concept is already illustrated in Figure 7, which directly compares observations from the different sensors used in the study and demonstrates the differences in spatial grain among WorldView-2 (1.6 m), PlanetScope (3 m), and Sentinel-2 (10 m). We believe that Figure 7 already provides the visual context requested by the reviewer and adequately conveys how the same landscape is represented at different spatial grains. Given that the manuscript is also at the journal's suggested limit for display items, we have therefore retained the existing figures.

L205: "Imagery was selected to align as closely as possible with field and drone surveys while minimising phenological mismatch" -> there are two dates given in the supp for 2x sensors, May and Sept, please explain why?

Thank you for identifying this error. The dates reported in the Supplementary Materials were mistakenly taken from the downloaded imagery archive rather than the imagery used in the analyses. The supplementary table S4 has been corrected to include only the acquisition dates of the imagery used in this study.

L208: what justification for these indices?

We have revised the text to provide a concise rationale for our choice of vegetation indices, supported by relevant literature, while avoiding unnecessary expansion of the methodological discussion. The revised sentence now reads: "Pre-processing included sensor-specific radiometric and atmospheric correction, after which predictor layers were assembled from spectral bands and vegetation indices (NDVI, SAVI, MSAVI2 and MTVI2; Table S5), selected to capture complementary variation in vegetation greenness and canopy signal under sparse vegetation and exposed-soil conditions (Rajah et al. 2019; Rebelo et al. 2021b; Francis et al. 2024; Slade et al. 2023)."

Section 2.5 opening paragraph seems repetition of 2.1

Thank you for highlighting this. We have extensively revised Section 2.1 to reduce unnecessary detail and repetition with the subsequent methods sections, while retaining a concise overview of the overall experimental design to provide context for the more detailed description of the multi-scale analytical framework in Section 2.5.

L239-242: please show the spatial blocking in your supp data. Which study areas were blocked, and which were included?

L243 onwards: also the canopy height model? Some kind of conceptual framework would be helpful here for authors (i.e. not Fig S2; the study design not the workflow).

Thank you for this suggestion. We have considered several options for presenting a conceptual framework for the study design, including approaches that incorporate the canopy height model, but have not identified a visual framework that we feel would add sufficient value beyond the existing figures and the description of the hierarchical multi-scale study design in Sections 2.1 and 2.5. We therefore prefer not to add a separate conceptual figure, as we feel this risks introducing an additional layer of visual complexity without materially improving the clarity of the study design.

L253: did you output an uncertainty layer with your classification results, and can you please include it? E.g. in supp if not in results.

Thank you for this suggestion. We did not generate a per-pixel uncertainty layer as part of the classification workflow, so we are unable to provide one from the existing analysis. We have instead strengthened the Discussion to acknowledge this limitation and to highlight probabilistic or fractional representations, including approaches such as conformal prediction (Angelopoulos and Bates 2022), as promising directions for future work to better quantify and communicate uncertainty in multi-scale classification products.

257-259: this is good, we would call this groundtruthing

We appreciate the reviewer's suggestion; however, we have chosen not to use the term "ground truthing." As argued by Woodhouse (2021), the term implies that field observations represent an absolute truth, whereas both field and remotely sensed measurements are subject to uncertainty, scale effects, and measurement error. In our study, the field data serve as reference or validation observations rather than unquestionable truth. We therefore retain terminology such as "field validation" and "in situ observations," which we believe more accurately reflects the role of these data.

L272: if you say: "maps", then you should have ranges of km² and % invaded, whereas abstract reports absolutes. I would suggest adding the ranges.

L275: another way to do this would be to explicitly collect data on different phases and include this in training data, rather than split these up using this rather cursory technique after the fact.

We agree that alternative methodological approaches are possible. The approach suggested by the reviewer would be appropriate for explicitly modelling invasion phase as a response variable, but this was not the objective of our study. Our aim was to quantify the spatial distribution and prevalence of *Neltuma* and, in addressing our fourth research question, use these mapped patterns to provide insights into invasion dynamics at the landscape scale. We have therefore retained the post-classification approach, with phases derived from mapped prevalence thresholds and used as a pragmatic, management-oriented interpretation of invasion status, rather than as directly modelled classes.

L278: how? Regression analysis? Given that the maps produced are models, and not observed data, you can run into some serious violations of assumptions. Please see the work by Behrer 2025 on Prediction-powered inference (PPI) on how to correct/account for this.

Thank you for highlighting this. We agree that formal statistical inference based on model-derived spatial predictions would require careful consideration of prediction uncertainty and the

assumptions associated with inference from predicted rather than observed data. However, we did not undertake regression or other formal inferential analyses of these spatial relationships. Rather, Figures S10 and S11 present exploratory spatial visualisations that we used to examine broad patterns in mapped vegetation cover in relation to proximity to settlements and roads. We have therefore revised the manuscript to clarify that these are descriptive observations used to inform interpretation of invasion dynamics and potential management relevance, rather than statistically tested relationships or formal predictions of future invasion. The revised text reads: *"We used the mapped invasion patterns to explore spatial variation in Neltuma occurrence and density in relation to settlement proximity, geomorphic setting and vegetation structure. These exploratory observations, informed interpretation of invasion dynamics and potential management relevance across the study system."*

Results

Figure 3 and 8: Rhigosum should be spelled 'Rhigozum'

Thank you for noting this typographical error. We have corrected *Rhigosum* to *Rhigozum* in Figures 3 and 8.

Figure 4: I had expected more of a nested design... this links to my comments in the methods around showing your study design conception framework. I had expected to see the different classifiers (you call learners) for each of the treatments, not separately.

The experimental design was not intended as a fully nested factorial design in which each learner was evaluated independently within every combination of inputs. Rather, the learners were compared separately within each modelling stage to assess their relative performance, while the treatment comparisons addressed the effects of different predictor and data configurations. We use the term "learner" consistent with terminology from the machine learning community and the mlr3 framework on which our analytical pipeline was built.

L294: some kind of contextual information may have helped with this, e.g. object based techniques like Simple Non-Iterative Clustering.

Thank you for this helpful suggestion. We have revised the Discussion (Section 4.1) to explicitly acknowledge that the pixel-based framework adopted here is expected to include some misclassification associated with canopy margins, mixed pixels and shadowed areas, and that these may be reduced through object-based approaches incorporating spatial context. We now identify these methods as a possible direction for future methodological refinement.

We have nevertheless retained the pixel-based approach because our objective was to evaluate the capacity of multispectral imagery acquired at different spatial scales to detect and characterise Neltuma invasion across a large landscape using a consistent analytical framework. The pixel-based workflow achieved high classification accuracy and strong plant-level detection performance, indicating that object-based segmentation was not required to meet the objectives of this study, while providing a simple and transferable framework for cross-scale comparison among sensors.

L298: "The multispectral + CHM + NDVI predictor stack" this kind of writing is very long winded and easily misunderstood. Couldn't you name your different treatments in a table, and then refer to these? This also links back to clearly communicating your study design.

We recognise that there are different approaches to communicating complex experimental designs. We have retained the explicit descriptions of the predictor stacks for consistency across the figures, methods and results, allowing readers to immediately see which data and predictor combinations were used without repeatedly referring to a separate lookup table. Given that these descriptions relate to a relatively small number of treatments, we consider this approach clearer and more transparent for the reader than introducing an additional naming system that would require translation between treatment labels and predictor configurations.

"Comparison of drone- and WorldView-2-based classifications also revealed biases not captured by conventional validation metrics" -> what does this mean? The table cited (S9) is just a confusion matrix. This is very conventional...

Thank you for this important observation. We agree that our original wording did not clearly distinguish between the satellite model's internal validation statistics and the discrepancies revealed through comparison with the drone-derived classification. We have revised Section 3.2 to make this distinction explicit and to communicate more clearly that independent comparison with the drone-derived classification identified additional errors that were not fully reflected by the internal validation metrics. The revised text now reads: *"Independent comparison with the drone-derived classification revealed discrepancies that were not fully reflected by the satellite classification's internal validation metrics. The satellite classification overpredicted Neltuma extent by 24.8% relative to drone classifications (Table S10), while 34.6% of Neltuma plants identified in drone data were liable to be classified as V. erioloba or other woody vegetation in satellite imagery. These comparisons indicate greater classification error than suggested by internal training and validation statistics alone, while reinforcing both the value of drone-informed calibration and the importance of spatially explicit validation."*

Table S9: please give the results, also for UA and PA, not just the raw table.

Figure 6. I can clearly see that Neltuma is closely associated with rivers (and/or groundwater interacting with those riverbeds).

Thank you for this observation. We agree that the mapped distribution shows a strong spatial association with the dry river systems. However, as discussed in our response to Comment L384, we do not consider this association alone sufficient to infer the underlying dispersal mechanism. In this landscape, roads, settlements and livestock infrastructure are concentrated along the ephemeral riverbeds, resulting in substantial spatial covariance among these features. We have therefore

retained a cautious interpretation, describing the observed pattern as consistent with livestock-mediated dispersal rather than evidence of causation, in both the revised manuscript and supplementary figure captions for S10 and S11.

The meaning of the x- and y-axis values in Figure 6C and Figures 8A and 8B should be clarified in the figure captions or removed if unnecessary

Figure 7: powerful figure... perhaps a more fair way to handle the S2 data would be where uncertainty is incorporated, or fractional cover. Whereby instead of forcing each pixel to assign a class, instead you assign a probability. E.g. see Angelopoulos & Bates 2022 -A Gentle Introduction to Conformal Prediction.

We agree that incorporating uncertainty or fractional cover would provide a more informative representation of sparse invasion patterns than assigning each coarse pixel to a single discrete class. The original Discussion already identified probabilistic or fractional representations as a potential direction for future multi-scale invasion mapping, but we have revised Section 4.2 to make this opportunity for methodological development more explicit and to more clearly articulate its relevance to uncertainty and spatial heterogeneity across observation scales. We also now signpost conformal prediction approaches (Angelopoulos and Bates 2022) as one potential framework for quantifying predictive uncertainty. We thank the reviewer for highlighting this point, which has helped us clarify and strengthen our discussion of how future approaches could build on the present work to provide more nuanced representations of sparse invasion patterns.

Figure 8: Neltuma should be italicized in the figure caption.

We have corrected this here and throughout the revised manuscript.

L384: I wouldn't be so sure. It looks like it has invaded all along the river. Correlation doesn't imply causation. You haven't considered all possible covariates. Given the critical importance of distance from river (from eyeballing the map), can you please update your analysis to include this?

Thank you for this comment. We agree that correlation does not imply causation, and we have not claimed a causal relationship. Our original wording carefully stated that the observed spatial distribution is "*consistent with field observations implicating livestock as a key vector of Neltuma spread*," rather than claiming proof of livestock-mediated dispersal. We do not consider water-mediated dispersal to be a likely primary mechanism in this system. The river channels are ephemeral, with several having experienced little or no surface flow for many years (due to upstream runoff harvesting, groundwater abstraction, and long-term precipitation patterns).

While distance to rivers could be considered as an additional covariate, we do not believe this would meaningfully resolve the attribution problem because roads, settlements, and livestock infrastructure are concentrated along the dry riverbeds, resulting in strong spatial covariance among these features. We therefore maintain that the observed pattern is consistent with, but does not prove, livestock-mediated dispersal. To help remind readers of this nuance and support informed interpretation, we have amended the caption of Figure S12 to read “*Fractional vegetation cover with proximity to roads, which are predominantly located adjacent to dry riverbeds.*”

Groundwater would also be great to include, but I would assume there are no data, and also that processing GRACE might be out of the scope of this study. I think it would also be helpful in your methods to explain how these covariates were selected, and why others were excluded. Was this done systematically? The methods should be transparent.

Thank you for this suggestion. As outlined in our research questions, this study was not designed to provide causal attribution of *Neltuma* spread or to comprehensively evaluate all potential environmental drivers. Rather, our objective was to quantify and map patterns of invasion using spatially explicit remote sensing products. Consequently, we use appropriately cautious language throughout the Discussion when considering possible mechanisms underlying the observed spatial patterns. Groundwater availability may potentially influence *Neltuma* distribution in arid systems; however, suitable groundwater observations are extremely limited within the study area, and incorporating satellite gravimetry products such as the Gravity Recovery and Climate Experiment (GRACE) mission would require substantial additional processing and interpretation beyond the scope of the present study.

Discussion

L396: *Neltuma* should be italicized

We have corrected this here and throughout the revised manuscript.

L432: but how feasible is this in terms of effort and money? I assume it is extremely expensive to do this drone section, and requires very specific expertise. So its nice, but what is the benefit to cost ratio?

Thank you for this point. We agree that the relative effort and cost of incorporating UAV observations are important considerations, but a comprehensive benefit-to-cost assessment is beyond the scope of this study. Moreover, such assessments are likely to vary substantially among applications, and geographic locations, and in our extensive experience change rapidly as UAV platforms, sensors and associated processing tools become more widely available and costs evolve. Rather than prescribing a universal benefit-to-cost threshold, we have sought to provide evidence that can help researchers and practitioners assess the relative advantages and limitations of fine-resolution UAV observations for a specific application. We have revised the discussion to better acknowledge these practical considerations and to emphasise that the value of UAV integration should be evaluated in relation to the ecological target, required spatial scale, available alternatives and intended management application.

L462: drone is not the only way to achieve this, there are other techniques as well (that are more cost-effective, but require expert ecological knowledge).

Thank you for this point. We agree that other approaches can provide complementary and potentially more cost-effective sources of fine-resolution ecological information. However, for use in similar cross-scale calibration and validation pipelines, these approaches still require accurately georeferenced observations, whether collected using drones or survey-grade GNSS instruments. We have therefore revised the text in this subsection to better clarify the specific role of precisely geolocated fine-resolution observations in our framework, without moving too far from the core narrative of this study.

L467: yes minimum mapping unit is an important consideration. This is a long section that confirms this basic remote sensing principle. Perhaps rather focus on how your work sought to overcome this.

We agree that these are foundational remote sensing concepts. However, given persistent confusion around the interpretation and application of scale and spatial resolution in some recent work, we felt it was important to retain a clear discussion of these issues in the context of our results. We have made minor revisions to Section 4.3 to place greater emphasis on the practical implications of addressing scale mismatch, including the value of fine-resolution observations and the need for cost-effective, scalable monitoring approaches. We have also integrated additional literature from related studies in eastern Africa to better contextualise our findings while retaining the core discussion of sensor grain and canopy structure as a key empirical finding of this study.

L496: but also cost effectively.

Thank you for this helpful suggestion. We strongly agree that cost-effectiveness is an important consideration for operational invasion monitoring, and have revised the paragraph to explicitly emphasise the need for cost-effective and scalable modelling and monitoring approaches.

Section 4.4: please update this based on my inputs

We have revised section 4.4 in light of earlier constructive suggestions.

L520 shifts naturally to interventions. Consider making this a section. Also address integrated approaches, and also consider why it was brought in initially (e.g. fire wood?)

Thank you for this suggestion. We agree that the discussion of management interventions follows naturally from the prognosis of continued expansion and densification. However, we felt that separating this into a standalone section would weaken the narrative, as the likely trajectory of the

invasion and the implications for management are closely linked. We have therefore retained these points within Section 4.4, but substantially refined the section by separating the discussion into clearer paragraphs, strengthening the articulation of the invasion trajectory and management implications, and integrating additional literature on invasive species monitoring and management from eastern and southern Africa. The revised section now also more explicitly considers integrated approaches combining early detection, targeted control, utilisation where appropriate, and community participation, while recognising the spatially differentiated management needs of sparse versus established infestations. We have not expanded the discussion of the original motivations for introducing *Neltuma* or subsequent control measures, as these are outside the primary focus of this study and are being examined in greater depth through a linked follow-up study involving a survey of approximately 200 people across the study area. This work will explore local perspectives on *Neltuma* in greater depth, including nuanced perceptions of its ecosystem service benefits and disbenefits, and will help inform understanding of the social dimensions and potential trade-offs associated with future management interventions.

L506 mentions a date but not a reason, as social resistance to totally removing it may be another challenge in its control.

Thank you for this observation. We deliberately used the term “deliberate introduction” to distinguish the initial establishment of *Neltuma* from accidental introduction or subsequent unintentional spread, as the species was intentionally introduced for sand dune stabilisation in 1980. We agree, however, that the social dimensions of *Neltuma* management, including perceptions of its benefits and potential resistance to complete removal, are important considerations that extend beyond the scope of the present study. We are currently undertaking a linked follow-up study involving a survey of approximately 200 people across the study area to examine these issues in greater depth, including nuanced perceptions of *Neltuma*'s ecosystem service benefits and disbenefits. This analysis is intended to help illuminate the potential social tensions and trade-offs that may shape responses to future control interventions. We therefore retain the wording to accurately describe the historical introduction while recognising that effective long-term management will need to account for both ecological impacts and local perspectives on the species.

Consider adding a section to the Discussion that addresses the potential transferability of this approach to other *Neltuma*-invaded regions, together with future recommendations. This would strengthen the manuscript's broader relevance.

Thank you for this suggestion. We agree that the broader transferability of the approach and implications for future mapping are important, and have strengthened these aspects throughout the revised Discussion rather than adding a separate section. In particular, Sections 4.2 and 4.3 now discuss how the hierarchical multi-scale framework, targeted UAV calibration and explicit consideration of sensor-grain limitations could be applied to other regions and to other sparse, heterogeneous invasion targets. Section 4.4 further considers the implications for spatially differentiated management and future monitoring of *Neltuma*. We have also revised the Conclusion to provide clearer recommendations for others mapping sparse invasions, including the use of targeted fine-resolution observations where they can materially improve landscape-scale inference while retaining operational scalability. We feel this integrated approach strengthens the manuscript's broader relevance without introducing a separate discussion section that would substantially repeat these points.

Conclusion

Could focus more clearly on recommendations for others wanting to map Neltuma, or this could form a section in the discussion

Thank you for this suggestion. We have addressed this point through substantial revisions throughout the manuscript, including integration of additional relevant literature and clearer articulation of the findings, their implications and recommendations for future mapping efforts. We consider that the revised Discussion now provides clearer guidance for researchers seeking to map Neltuma, particularly around the use of fine-resolution observations, cross-scale calibration, sensor grain, spatial validation and uncertainty. At the same time, we have sought to avoid presenting these recommendations as being applicable only to Neltuma, as many of the methodological insights arise from the broader challenge of detecting sparse, heterogeneous and spatially discontinuous ecological targets and are therefore transferable to other organisms with similar ecological characteristics. We feel this balance provides a more useful contribution to the wider Earth observation and invasion-mapping literature.

L540-541" delete

We have removed this sentence as suggested.

Consider deleting L546-548, this is commonly known, not a value-add of your study. I would limit conclusion to 1-2 sentences of your key findings.

We have removed this sentence as suggested.

L549-511 repetition, skip to result.

We have removed this sentence as suggested.

L556 onward: fine, but is this really the main value-add? Do people really see these systems as competing anyway? What about instead reflecting on the relative cost/benefits of drone incorporation for the additional accuracy % obtained? Is it worth it? Do you recommend it for others mapping sparse invasions?

Thank you for this suggestion. We agree that the key value of the approach is not simply that drones and satellites can be treated as complementary systems, but that fine-resolution observations can be strategically leveraged to improve the accuracy, robustness and management relevance of broader-scale monitoring. We have therefore substantially revised the Conclusion to make this more explicit and to provide clearer guidance for future mapping of sparse invasions. In particular, we now recommend incorporating UAV observations where the additional ecological detail and classification accuracy can materially improve landscape-scale inference, while using targeted acquisition to retain

operational scalability. This reflects our finding that UAV data can provide meaningful benefits for calibration and validation without requiring systematic fine-resolution coverage across the full monitoring area.

Supplementary Material

Rhigosum should be spelled 'Rhigozum': check this throughout the figures in the supplementary material

Thank you for spotting this typographical error. We have corrected *Rhigosum* to *Rhigozum* in Figures S1-S8 and S10 and S11 in the revised supplementary information.

Fig S12: did you include iNaturalist to look for observations, if not, please add these (there are a few: https://www.inaturalist.org/observations?nelat=-21.94631001953545&nelng=29.737596753070445&subview=map&swlat=-27.433413495640867&swlng=14.389696362445447&taxon_id=1493090)

Thank you for this helpful suggestion. We reviewed the available iNaturalist records to assess whether they could strengthen Figure S12. However, there appear to be only three observations within the western Botswana focal area, all of which coincide with locations already represented by our own author observations. Most of the apparently relevant iNaturalist records are located across the border in South Africa and therefore fall outside the focus of our Botswana study area. We have therefore not incorporated these records into Figure S12, as they do not add additional information within the area of interest.

Code

The Zenodo notebook link for the output classifications should be checked and corrected, because it is not working.

Thank you for identifying this oversight. The dataset had been uploaded to Zenodo but was not yet published, rendering the DOI inactive. The dataset has now been published and is accessible via the original DOI.

Reviewer 2

This manuscript “Detecting sparse biological invasions through multi-scale remote sensing: a drone-satellite framework for mapping Neltuma in dryland ecosystems” (ID: ERE-100330) takes the Kalahari rangelands in southwestern Botswana as the study area and investigates the use of multi-scale UAV and satellite remote sensing for detecting and mapping sparse invasive Neltuma. The topic has certain novelty and practical management relevance. The experimental design is systematic, employing multiple sensors and machine learning algorithms, and using relatively rigorous evaluation methods such as spatial block cross-validation. However, major revision is required before further consideration. My specific comments are in the attached file.

We thank Reviewer 2 for their positive assessment of our manuscript and for recognising its novelty, practical management relevance, and the strengths of the experimental design. We also appreciate the constructive feedback regarding the communication and interpretation of satellite classification performance. In response, we have revised the Abstract, Discussion and Conclusion to more clearly communicate the discrepancy between internal classification accuracy and independent UAV-based validation, and to better discuss the implications of these differences for operational mapping and management.

L343-346: The actual performance of satellite classification is substantially overestimated; the large discrepancy relative to UAV validation is insufficiently disclosed. On page 22, the authors clearly report that relative to UAV classification, the WorldView-2 satellite classification overpredicts Neltuma extent by 24.8%, and 34.6% of UAV-identified plants are liable to be misclassified as other woody vegetation. These two metrics indicate that real-world errors are substantially larger than the pixel-level accuracy (75.8%) suggests. However, in the Abstract, Conclusions, and management recommendations, the authors rely mainly on the higher 75.8% figure, mentioning the discrepancy only briefly in the Discussion. This undermines the credibility of the invasion extent estimates derived from the satellite map (e.g., “16 km² established, 356 km² expanding”). The authors should explicitly report these two discrepancy metrics in the Abstract and Conclusions, and discuss their concrete implications for management decisions (e.g., costs of false positives and false negatives in targeted removal).

Response:

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